

Environmental and Development Economics

Module 1 - Introduction

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Who am I?

- ▶ Raahil Madhok
 - ▶ Assistant Professor of Applied Economics
 - ▶ Started in 2023
- ▶ From Toronto, Canada
- ▶ Education
 - ▶ BA in Economics and Environmental Studies at McGill University (Montreal)
 - ▶ MA and PhD at University of British Columbia (Vancouver)
- ▶ Research interests:
 - ▶ environmental economics, development economics, political economy

Introduce yourself

- ▶ First, lets do introductions
- ▶ Name, year, memorable summer activity, **research interests**
- ▶ Why are you taking this class?

- ▶ **Class Time/Location:** Mon/Wed 8:45am-10:25pm, Ruttan B42
 - ▶ Last class: Oct 26th
- ▶ **Office Hours:** Tues/Thurs 11:30am-12:30pm, Ruttan 337D
- ▶ **Course website:** <https://github.com/rmadhok/enviro-dev-grad>
 - ▶ lectures, assignments, syllabus
 - ▶ Try to skim reading(s) beforehand
- ▶ **Assignments:** Upload through Canvas

Today

- ▶ Why study environmental economics in LMICs?
- ▶ Course overview + detailed outline
- ▶ Grade breakdown
- ▶ If time: conceptual framework for environment & development econ

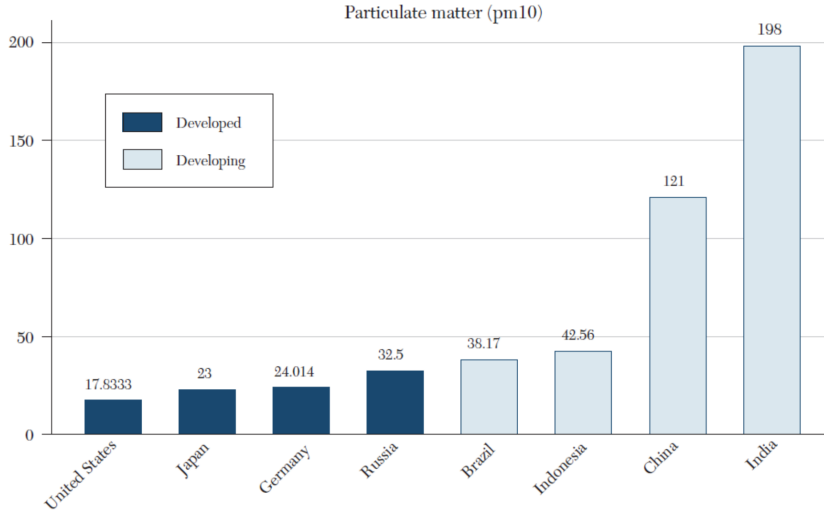
Why study environmental economics in LMICs?

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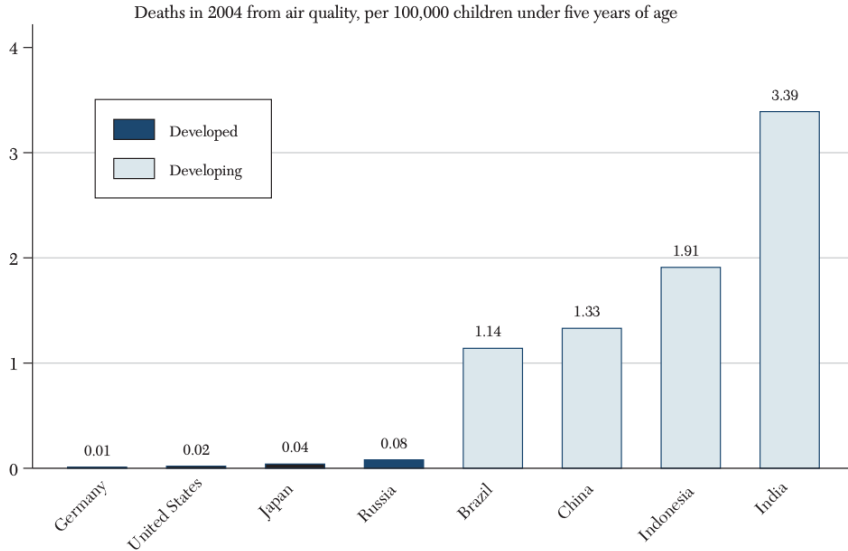
- ▶ Environmental quality is worse and has worse consequences in LMICs
 - ▶ Highest pollution, highest deforestation
 - ▶ Pressing need for conservation policy as LMICs grow/urbanize
- ▶ New field: room for applied theory, empirical innovation
- ▶ Data breakthroughs
 - ▶ Measurement: remote sensing, DHS, etc.
 - ▶ Access: lower barriers to government access and experimentation
- ▶ Evidence needed; big implications for poverty alleviation

Environmental quality worse in LMICs

Panel A. Air pollution



Disease burden higher in LMICs



Is environmental economics different in LMICs?

My answer: sometimes...

- ▶ Magnitudes
 - ▶ Same questions, but costs and benefits different
- ▶ **Local** environmental quality is more important
- ▶ Different topics
 - ▶ cookstoves, enforcement/corruption, ethnic favoritism
- ▶ Institutions and state capacity

Course Overview

Course Overview: There is no textbook

Instead, I am organizing around **FIVE** key questions:

- ① How does economic development affect the environment, and vice versa?
- ② Why is environmental quality so bad in developing countries?
- ③ What are the costs of poor environmental quality in developing countries?
- ④ Why is WTP for environmental quality low in developing countries?
- ⑤ What are the political economy barriers to environmental protection?

Course Approach

I will:

- ▶ Frame (almost) each topic with some theory
- ▶ Teach applied papers
 - ▶ research design, identification strategy, estimation techniques
 - ▶ aim for two per class (please skim beforehand)
- ▶ Emphasize recent papers

I will NOT:

- ▶ Teach econometrics
- ▶ Teach coding
- ▶ Teach every topic in environment/development

Course Goals

- ① Show you environment/development research frontier
- ② Inspire your thesis/JMP ideas
- ③ Advance your training as applied microeconomists
- ④ Show you what makes a top-tier research question

Course Structure

- ▶ This is a brand new class, so I give myself leeway to make changes
- ▶ You have the unique opportunity to determine direction of the course
 - ▶ Think about what topics do and don't interest you
 - ▶ And let me know!
- ▶ Please check the course website regularly for updates

Course Outline and Topics

Module 1: Introduction

- ▶ Lecture 1: Course intro + how to use theory to ask the right questions

Module 2: The effect of development on the environment

- ▶ Lecture 2: Income effects
- ▶ Lecture 3: Access to capital (technology and infrastructure)

Module 3: The effect of environment on development

- ▶ Lecture 4: Health
- ▶ Lecture 5: Productivity

Module 4: Why is WTP low in developing countries?

- ▶ Lecture 6: Revealed preference approaches
- ▶ Lecture 7: Incentive compatible approaches

Module 5: Environmental Policy Design

- ▶ Lecture 8: Monitoring, enforcement
- ▶ Lecture 9: Barriers to optimal design

Module 6: Political Economy of the Environment

- ▶ Lecture 10: Electoral cycles, corruption
- ▶ Lecture 11: State Capacity

Module 7: Research Proposal Presentations

- ▶ Lecture 12: TBD
- ▶ Lecture 13: Presentations
- ▶ Lecture 14: Presentations

Grade Breakdown

Breakdown

In-class presentations	10%
Replication Assignment	20%
Research Proposal	60 %
Participation	10%

In-class presentations (10%)

- ▶ I want you to become expert presenters after taking this class
- ▶ At start of **each** class, you'll give a 10 min paper presentation
 - ▶ The paper for presentation is on the syllabus
- ▶ You will submit **10** summary slides (5% of grade)
 - ▶ motivation, research question, methods, results
 - ▶ You may use AI
 - ▶ You can submit slides as a group of up to four
 - ▶ 10 mins presentation + 5 mins Q&A (5% of grade)
- ▶ I will select presenter on-the-spot
 - ▶ **randomly** with replacement**

** If you are never chosen, your grade is based on slides.

Problem Set (20%)

- ▶ You will replicate an environment/development paper
 - ▶ You will also **extend** the results
 - ▶ Many papers on the syllabus have replication files
- ▶ You will submit a write-up explaining what you did
- ▶ You will become familiar with coding in publication-quality papers
- ▶ You will use R or Stata
- ▶ Last time student's published their paper!

McWay, Ryan, and Matthew Braaksma. “The political consequences of resource scarcity: Targeted spending in a water-stressed democracy. A replication study[...].” *World Development Perspectives* 39 (2025): 100707.

Research Proposal (60%)

Written Proposal	30%	Oct. 30
First Draft	pass/fail %	Oct. 5th
Peer Review	20%	Oct. 14th
Proposal Presentation	10%	Oct 21/26

- ▶ You will develop a research proposal for an original idea
 - ▶ You are NOT expected to actually do the analysis
 - ▶ I will provide small deadlines (first draft, etc.) along the way
- ▶ Come to office hours to pitch your idea
- ▶ You will peer review each others proposals
- ▶ You will present the proposal at the end of the semester (20 mins)

Participation (10%)

- ▶ 7 points for attendance, 3 points for engagement
 - ▶ in class engagement, office hrs, emails
- ▶ Attendance based on tiered schedule:

Classes Attended	Classes Missed	Attendance Points
12–14	0–2	7/7
9–11	3–5	5/7
6–8	6–8	3/7
3–5	9–11	1/7
0–2	12–14	0/7
Engagement	–	+ 0–3
Total		10/10

Questions?

- ▶ **Guiding question:** Why is environmental quality so low in LMICs?
- ▶ Your explanations
- ▶ **Main goal:** Conceptual framework
 - ▶ Four *theory-informed explanations*
 - ▶ Set the stage for rest of class

Why is environmental quality low in LMICs?

- ▶ MWTP is low (paradox)
 - ▶ Berkouwer and Dean (2022): \$12 for clean air
 - ▶ Kremer et al. (2013): ~ \$4 for clean water
 - ▶ Imply VSL \$USD 860 vs \$USD 8.6 million for USA
- ▶ Do we take this as given? Perhaps status quo is optimal
 - ▶ is bad environmental quality another dimension of poverty?
- ▶ Is welfare loss from pollution greater in rich countries, even though they're cleaner?
- ▶ **What are your explanations?**

Theory-informed Explanations

Greenstone and Jack (2013)

Aside: why is applied theory important?

- ▶ Builds structure for answering big (and small) questions
- ▶ Generates potentially unexpected insights w/ testable predictions
- ▶ In reverse: helps rationalize results
- ▶ Gets you into better journals (and better jobs)
- ▶ Field is headed that way (from my recent experience)

Conceptual Framework of Environmental and Development Economics

- ▶ Social planner chooses optimal EQ where social $MWTP_e = MC$
 - ▶ Need to know MWTP for representative agent

Set up:

- ▶ n identical agents with utility from consumption, EQ, and health
- ▶ Initial individual wealth y_0 , health h_0 , environmental equality e_0
- ▶ health depends on self-protection, s , and e
- ▶ Assume perfect markets (i.e. no externalities) to benchmark first-best

- ▶ Agent chooses c , Δe , and s to maximize:

$$U(e, h(s, e), c) \quad \text{s.t.} \quad y \geq c_e(\Delta e) + c_s(s) + c$$

- ▶ where wealth (endowment + income) and experienced EQ are:

$$y = y_0 + \Delta y(e, h(s, e))$$

$$e = e_0 + \Delta e + a(c, s)$$

- ▶ where $a(c, s)$ captures impact of c and s on EQ

Model Particulars

- ▶ EQ affects utility directly through existence value
- ▶ EQ affects utility indirectly via health (which also affects income)
 - ▶ e.g. pollution exposure affects productivity
 - ▶ This can be mitigated by self-protection, s (e.g. mask, air purifier)
- ▶ EQ affects income, which in turn affects utility via budget constraint
 - ▶ e.g. agricultural productivity
- ▶ Experienced EQ depends directly on Δe , and indirectly via c , s
 - ▶ $a(c, s)$: defensive investments i.e. clean cookstove, bottled water, etc.

MWTP for improving environmental quality

- ▶ Let $\lambda_e = \frac{\partial u}{\partial \Delta e}$, $\lambda_y = \frac{\partial u}{\partial c}$
- ▶ Set up lagrangian and solve for $MWTP_e$:

$$MWTP_e = \frac{\lambda_e}{\lambda_y} = \frac{1}{\lambda_y} \left(\frac{\partial u}{\partial e} + \frac{\partial u}{\partial h} \frac{\partial h}{\partial e} \right) + \frac{\partial \Delta y}{\partial e} + \frac{\partial \Delta y}{\partial h} \frac{\partial h}{\partial e}$$

- ▶ aesthetic benefit from improved EQ (converted to dollars)
- ▶ indirect benefit of EQ for health (converted to dollars)
- ▶ direct impact of EQ on income and indirect impact via health

Note: if $U''(c) < 0$, low $y \rightarrow$ high MUC (λ_y) and low $MWTP_e$

The First Best Outcome

- ▶ Agent sets ratio of marginal costs = ratio of marginal benefits

$$MWTP_e = \frac{\partial c_e}{\partial \Delta e}$$

- ▶ For social planner to aggregate over n , we must assume:
 - ▶ No preferences of her own
 - ▶ Can observe true MWTP
 - ▶ Anything else?
- ▶ Do these hold in LMICs?

Course Structure

- ▶ Set the stage:
 - ▶ how does environment affect development ($\frac{\partial h}{\partial e}$) (module 2)
 - ▶ how does development affect the environment (module 3)
- ▶ Bulk of course:
 - ▶ Explain why environmental quality low in LMICs
 - ▶ Identify as many parameters of the social planner problem as possible
- ▶ Goal: where can you make a contribution?
 - ▶ E.g. Lots of work on $\frac{\partial h}{\partial e}$, but very little on $\frac{\partial \Delta y}{\partial e}$

Why is environmental quality so low in LMICs?

Four explanations informed by the model:

- ① High marginal utility of consumption (efficient markets)
- ② High marginal abatement costs (efficient markets)
- ③ Political economy distortions (first best violation)
- ④ Market failures (first best violation)
 - ▶ frictions cause revealed MWTP $\neq$ true MWTP

Preview of Answers

1. High marginal utility of consumption

- ▶ Intuitively, poor people care more about meeting basic consumption needs
- ▶ Economically, agent trades off c and e by setting $u'(c) = u'(e)$
 - ▶ If $u''(c) < 0$, prefer c at lower levels of y
 - ▶ even if health benefits of e are large!
- ▶ **Very** few revealed preference studies on $MWTP_e$
 - ▶ Kremer et al. (2013) randomly clean up springs in Kenya
 - ▶ WTP USD 11/year for clean water; VSL of USD 860
 - ▶ In progress: MWTP for biodiversity (Braaksma, Lasway, Madhok and McWay, 2026)
- ▶ Larger literature on $u'(h)$ also suggests low valuation (Berkouwer and Dean, 2022)

2. High MC

- ▶ High MAC suggests sub-optimal environmental quality. Why?
 - ▶ Upward sloping MAC suggests low MC in poor countries
- ▶ MC not only driven by MAC; also reflects weak state capacity
 - ▶ Enforcement (Duflo et al., 2013)
 - ▶ Incentives (Gulzaar and Dipoppa, 2024)
- ▶ High MC **does not** mean deviation from first best

3. Political economy

- ▶ Social planner includes own utility weights social welfare function
 - ▶ i.e. corruption
- ▶ Many examples from LMICs
 - ▶ pollution (Duflo et al., 2013)
 - ▶ deforestation (Burgess et al., 2012)
 - ▶ human-wildlife conflict (Madhok et al., 2026)
- ▶ Leads to second best policy (inefficient)

4. Market Failures

- ▶ This is partially a course on development economics
 - ▶ About market failures: land, labor, credit, etc.
- ▶ Implication for us: revealed $MWTP_e \neq$ first best $MWTP_e$
- ▶ Example: weak property rights $\rightarrow$ underinvestment in e
 - ▶ Underestimate $MWTP_e$ from observed data
 - ▶ RCT evidence from crop-burning PES contracts: Jack et al. (2024)

Lots of room for research

- ▶ Environment and development economics is new
 - ▶ Challenge: find something unique about LMICs
- ▶ Goal: identify model parameters
- ▶ Evidence on many parameters are absent
- ▶ Barriers to research in LMICs are falling
 - ▶ remote sensing, administrative/survey data, webscraping

Next Class

- ▶ In-class presentations
- ▶ Impact of economic development on the environment (income effects)
- ▶ Impact of development on the environment (forests, biodiversity)

Aside: Best Practice for Short Presentations

Best Practices: Structure

- ▶ You cannot present a paper in 10 minutes
 - ▶ Do not give detailed lit review or go into extreme detail
- ▶ Instead, you are giving a trailer for the movie
 - ▶ Convince the audience that they should read the paper
- ▶ Your goal is only to state why the paper is important, and what you did
- ▶ Convey paper's importance in first slide

Best Practices: Slides

- ▶ Motivation + broad research question (1 slide)
- ▶ Full paper overview (1 slide)
- ▶ Lit review (optional, 1 slide)
- ▶ Background (1 slide)
- ▶ Data (2 slides)
- ▶ Empirical Strategy (2 slides)
- ▶ Results (2 slides)
- ▶ Summary (1 slide)

Example of Preview Slide

- ▶ **Question:** How do firms react to tribal forest policy?
- ▶ **Idea:** Model aggregate economic response and changes in firm composition
- ▶ **Setting:** India Forest Rights Act (2008)
 - ▶ Imposes transaction cost on firms
- ▶ **Data:** Manufacturing census (2001-2015); Deforestation permits (2001-2021)
- ▶ **Empirical Strategy:** Diff-in-diff using policy shift in tribal and non-tribal district

Results Preview

- 1) decline in firm activity, 2) less forest encroachment by industry
- 2) larger, but less productive firms survive

Best Practices: Slide Format

- ▶ 1 minute per slide
- ▶ Avoid chart junk
- ▶ One line bullets
 - ▶ No need for full sentences
- ▶ Vary text slides and text + image slides
- ▶ Don't put too many equations
 - ▶ Save details for speaking, or talk “about” equation
- ▶ Summarize findings again at the end

Best Practices: Presentation

- ▶ Speak clearly and loudly
- ▶ Speak slowly
- ▶ Look at audience; Do not show your back
- ▶ Do not stand in front of slides
- ▶ Avoid jargon
- ▶ Avoid pacing around room
- ▶ Stick to your time limit